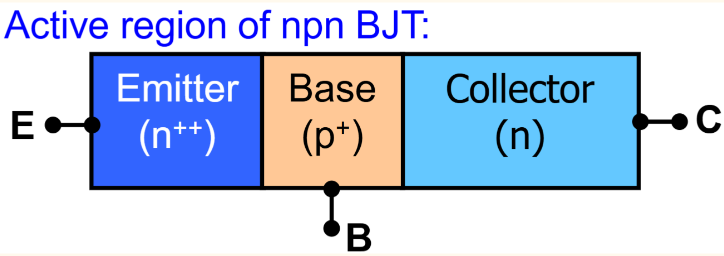
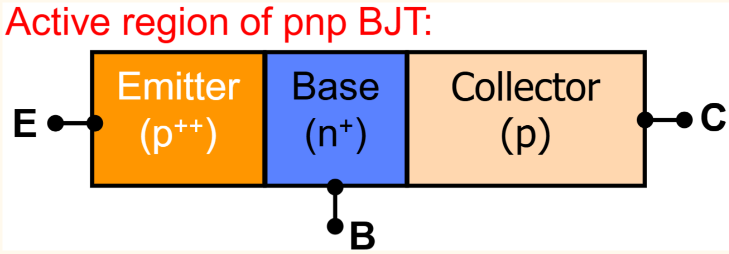
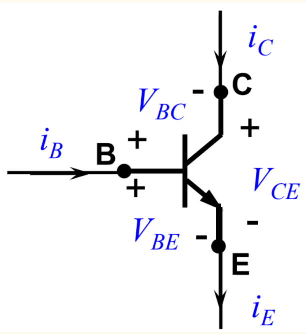
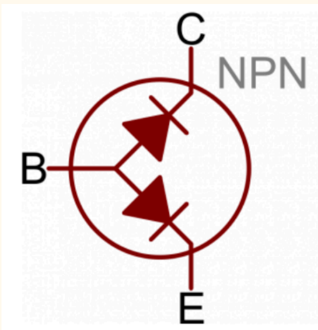
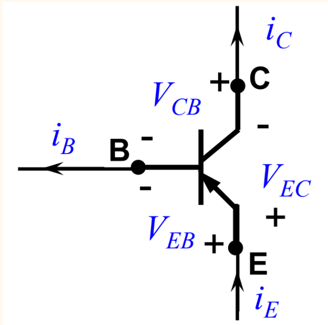
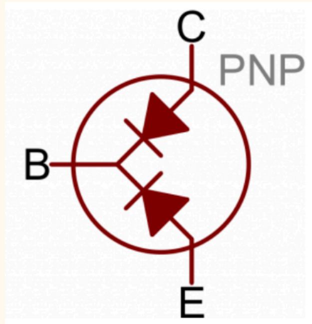
BJT Summary

Operating Modes

npn	pnp
	
 	 

npn

Mode of Operation	Emitter-Base Junction	Collector-Base Junction	Application
Cut-off	Reverse biased (V_{BE} < 0 for npn)	Reverse biased (V_{BC} < 0 for npn)	Logic - OFF State
Forward Active	Forward biased (V_{BE} > 0 for npn)	Reverse biased (V_{BC} < 0 for npn)	Amplifier
Saturation	Forward biased (V_{BE} > 0 for npn)	Forward biased (V_{BC} >0 for npn)	Logic - ON State
Reverse Active	Reverse Biased (V_{BE} < 0 for npn)	Forward Biased (V_{BC} > 0 for npn)	Not used

pnp

Mode of Operation	Emitter-Base Junction	Collector-Base Junction	Applications
Cut-off	Reverse biased (V_{EB} < 0 for pnp)	Reverse biased (V_{CB} < 0 for pnp)	Logic- OFF State
Forward Active	Forward biased (V_{EB} > 0 for pnp)	Reverse biased (V_{CB} < 0 for pnp)	Amplifier
Saturation	Forward biased (V_{EB} > 0 for pnp)	Forward biased (V_{CB} >0 for pnp)	Logic - ON State
Reverse Active	Reverse Biased (V_{EB} < 0 for pnp)	Forward Biased (V_{CB} > 0 for pnp)	Not used

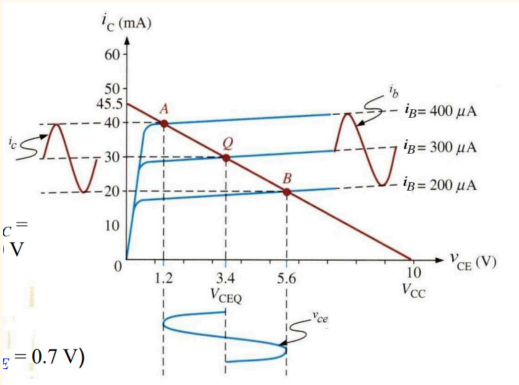
Forward Active IV Characteristics

Without Early Effect	With Early Effect (V_A ~ 100-200V)
$i_C = I_S e^{v_{BE}/V_T}$ $i_B = \frac{I_S}{\beta} e^{v_{BE}/V_T}$	$i_C = I_S e^{v_{BE}/V_T}(1 + \frac{v_{CE}}{V_A})$ $i_B = \frac{I_S}{\beta} e^{v_{BE}/V_T}(1 + \frac{v_{CE}}{V_A})$

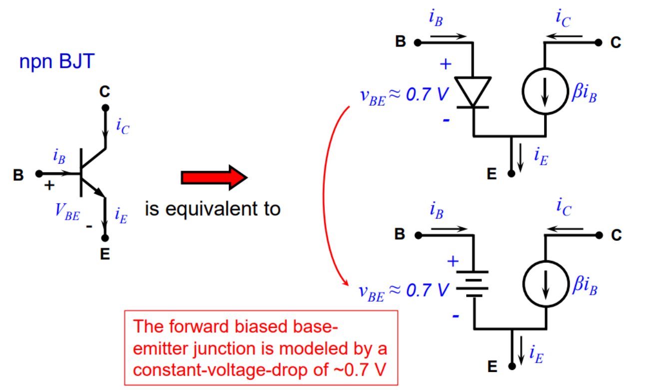
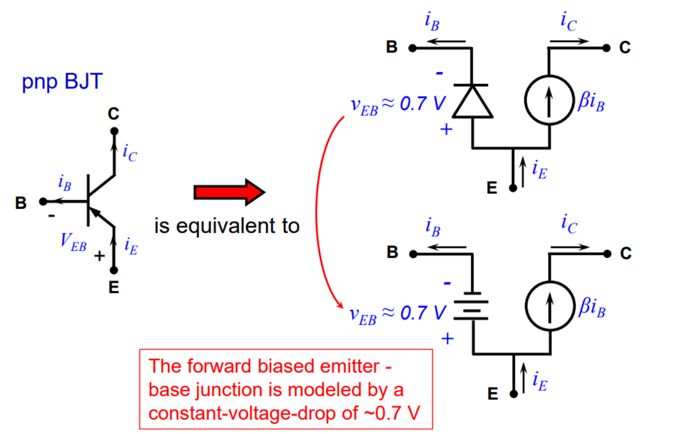
$i_E = i_C + i_B \approx i_C$	$i_C = \beta\, i_B$	$i_E = (\beta + 1)\, i_B$	$i_E = (\frac{1}{\beta} + 1)\, i_C$

α = β/(1+β)

Modelling



DC

npn	pnp
As the emitter-base junction is forward biased, $v_{BE} \approx 0.7$ V, it can be modeled by a constant voltage source (using the constant-voltage-drop model for a forward biased pn-junction). 	Large-signal model of a pnp BJT: polarities of voltages and directions of currents are opposite to those of an npn BJT. 

- direction of current source and voltage drop is opposite

Thevenin Equivalent Method

1. Check npn or pnp
2. Assumption:
   a. BJT is under forward active mode ⇒ $I_C = \beta I_B$
3. Model base biassing circuit by Thevenin equivalent
4. With the Thevenin equivalent, find base current
5. With base current and β find collector current
6. With collector current find emitter current and collector voltage
7. Check operating mode to verify assumption
   a. for npn: V_{BE} > 0.7 V, emitter-base is forward biassed & V_{BC} < 0 V, base-collector is reverse biassed
   b. for pnp: V_{EB} > 0.7 V, emitter-base is forward biassed & V_{CB} < 0 V, base-collector is reverse biassed

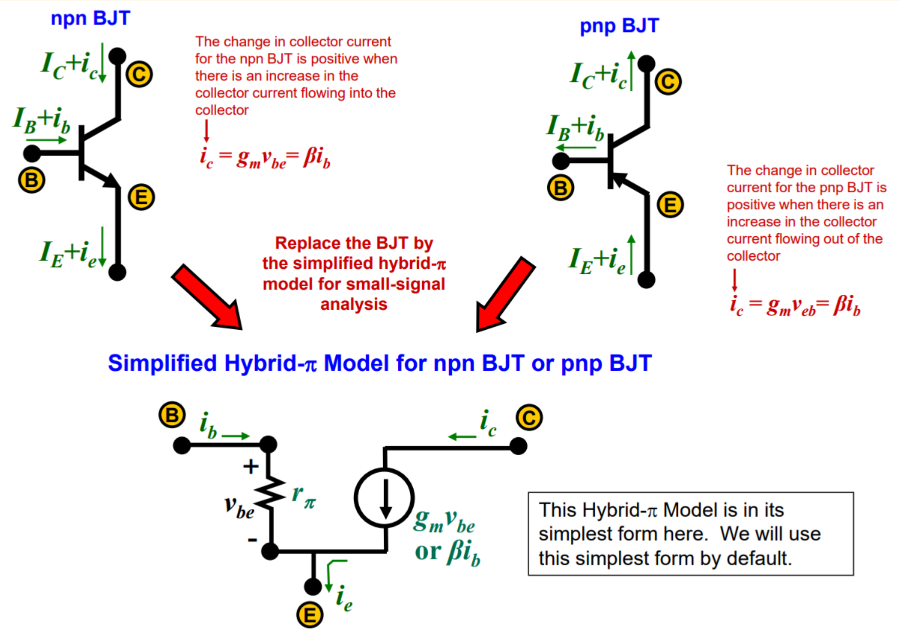
Voltage divider method

1. Check npn or pnp
2. Assumption:
   a. BJT is under forward active mode ⇒ $I_C = \beta I_B$
   b. Biassing circuit current >> Base current ⇒ base voltage can be easily found
3. From potential divider, find base voltage
4. From base voltage and 0.7 V drop, find emitter voltage
5. From emitter voltage and emitter resistor, find emitter current
6. From emitter current and β, find collector current and base current
7. From collector current and collector resistor find collector voltage
8. Check base current and biassing circuit current to verify assumption 2
   a. if biassing circuit current is not 10x of base current, redo analysis with Thevenin method
9. Check operating mode to verify assumption 1

a. for npn: $V_{BE} > 0.7$ V, emitter-base is forward biassed & $V_{BC} < 0$ V, base-collector is reverse biassed
b. for pnp: $V_{EB} > 0.7$ V, emitter-base is forward biassed & $V_{CB} < 0$ V, base-collector is reverse biassed

AC

Simplified Hybrid-π Model



$$r_\pi = r_i$$

EE2027 used r_π
EE5507 used r_i

$$\beta = g_m r_\pi$$

※ Its easy to remember this
※ Whever the input current is, it goes into r_π and generate a voltage
※ Output current is gepending on the voltage via g_m
※ So, the overall current gain is simply $g_m r_\pi$

No Early Effect

$$i_C = I_S e^{v_{BE}/V_T}$$

Parameters

$$g_m = Transconductance$$

$$g_m = \frac{I_C}{V_T}$$

$$r_\pi = input\ resistance$$

$$r_\pi = \frac{\beta V_T}{I_C}$$

$$\beta = g_m r_\pi$$

$$r_\pi = \frac{\beta}{g_m}$$

$$g_m = \frac{\beta}{r_\pi}$$

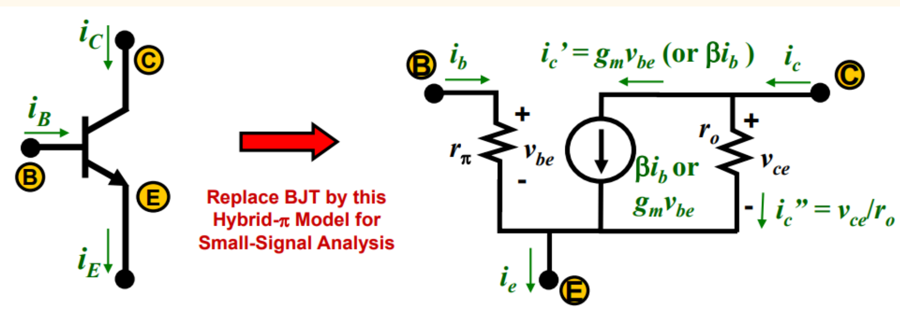
Relationships

$$i_c = g_m v_{be}$$

$$v_{be} = r_\pi i_b$$

$$i_c = \beta i_b$$

Full Hybrid-π Model with Early Effect



Early Effect

$$i_C = I_S e^{v_{BE}/V_T}\left(1 + \frac{v_{CE}}{V_A}\right)$$

Additional Parameter

$$r_o = output\ resistance$$

$$r_o = \frac{V_A}{I_C}$$

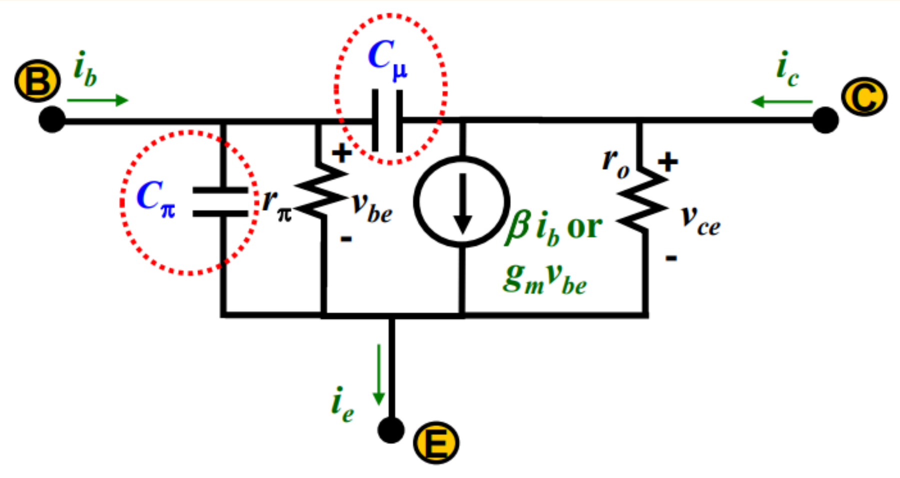
$$r_o = \infty\ if\ V_A\ not\ given$$

Relationships

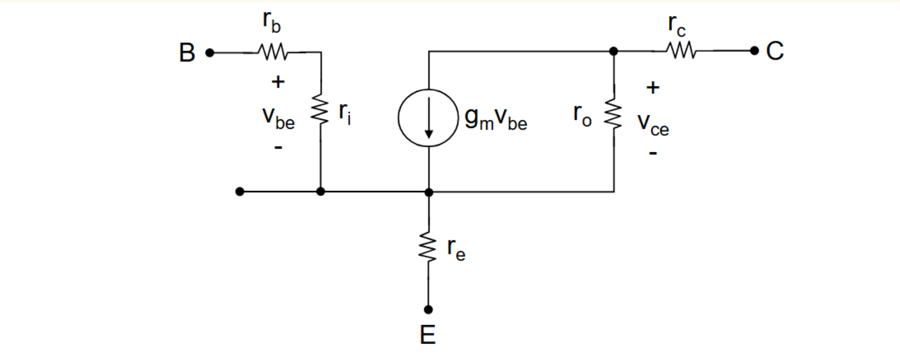
$$i_c = g_m v_{be} + \frac{1}{r_0} v_{ce}$$

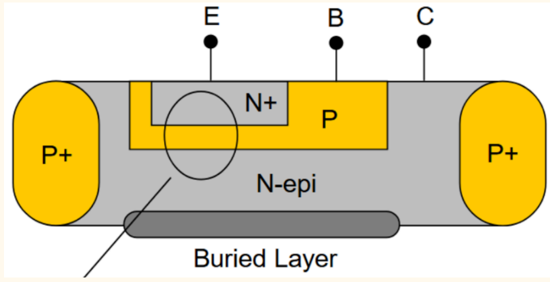
$$i_c = \beta i_b + \frac{1}{r_0} v_{ce}$$

Full Hybrid-π Model at High Frequency (EE2027)



Full Hybrid-π Model at Low Frequency (EE5507)





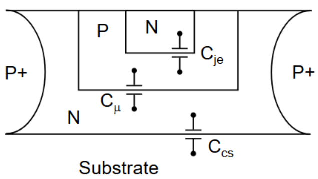
r_e is small, because the emitter is on top and lightly dope

r_c is non negligible without the buried layer

r_b can be minimised with fingers, like Rg in MOSFET, see [here](#)

Full Hybrid-π Model at High Frequency (EE5507)

- Junction Capacitance (C_{je}, C_μ, C_{cs})

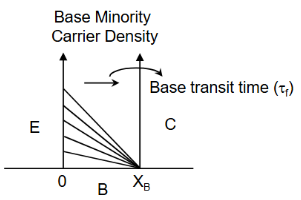


$$C_j = \frac{C_{jo}}{\sqrt[m]{1+\frac{V_{Bias}}{\phi_{bi}}}} \qquad m \approx \frac{1}{3} \sim \frac{1}{2}$$

C_{je} : forward bias

C_μ, C_{cs} : reverse bias

- Base Charging Capacitance (C_b)



$$\Delta Q_n = i_c \cdot \tau_F$$

$$\Delta Q_h = C_b \cdot v_{be}$$

$$\Delta Q_n = \Delta Q_h \longleftarrow \text{Charge Neutrality}$$

$$\Rightarrow C_b = \tau_F \cdot \frac{i_c}{v_{be}} = \tau_F \cdot g_m$$

$$\approx 250\,ps \cdot \frac{1}{26\Omega} = 10\,pF$$

Diffusion = Base Charging capacitance

⋙ In NPN, base is forward bias then a lot electron injected into base then get sweep to the collector.

⋙ Base terminal will have +ve charge, then the negative electron is in the base region.

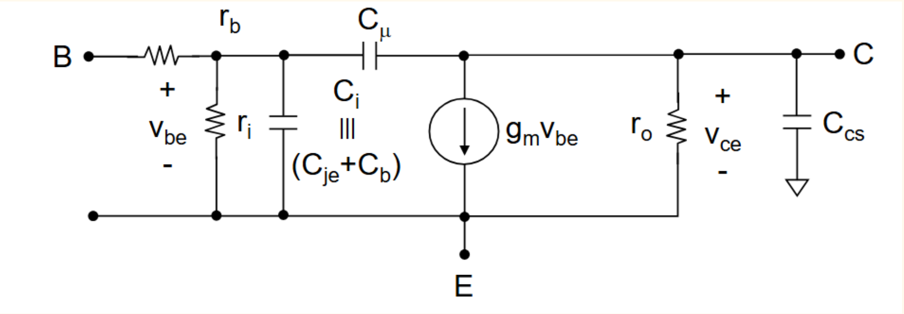
⋙ Turning BJT off will not off instantly, it take time for electrons to move from base region and get collected by the collector.

⋙ This is called transit time, τ_F

⋙ Can deduce the diffusion capacitance from transit time

⋙ Dominant factor for the BJT speed

make the E to C width thinner = base width thinner, then BJT faster

<table>
<tr><td>

$C_b >> C_{je}$</td><td></td></tr>
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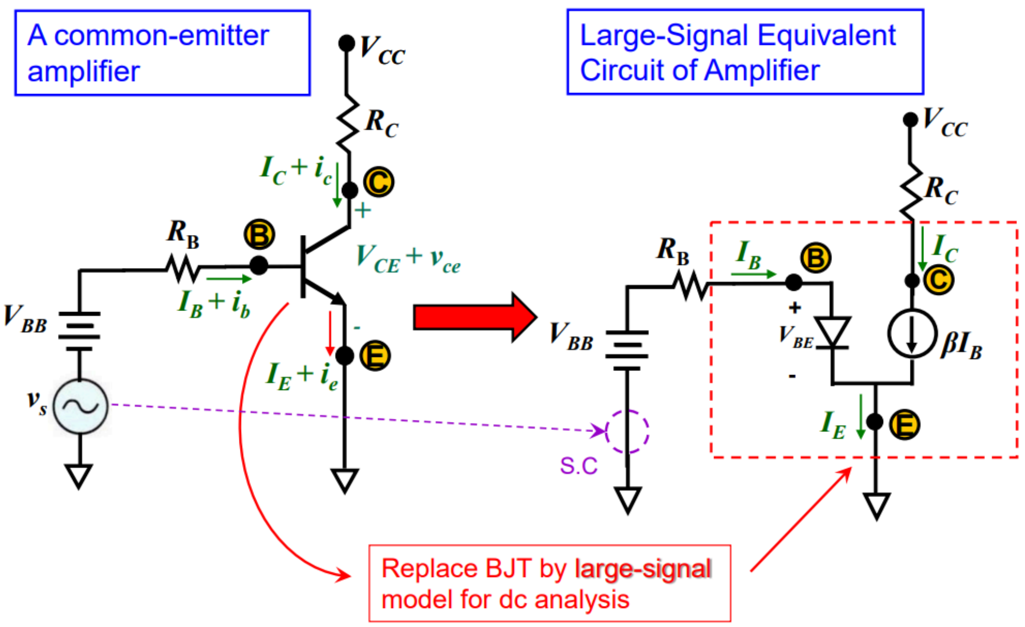
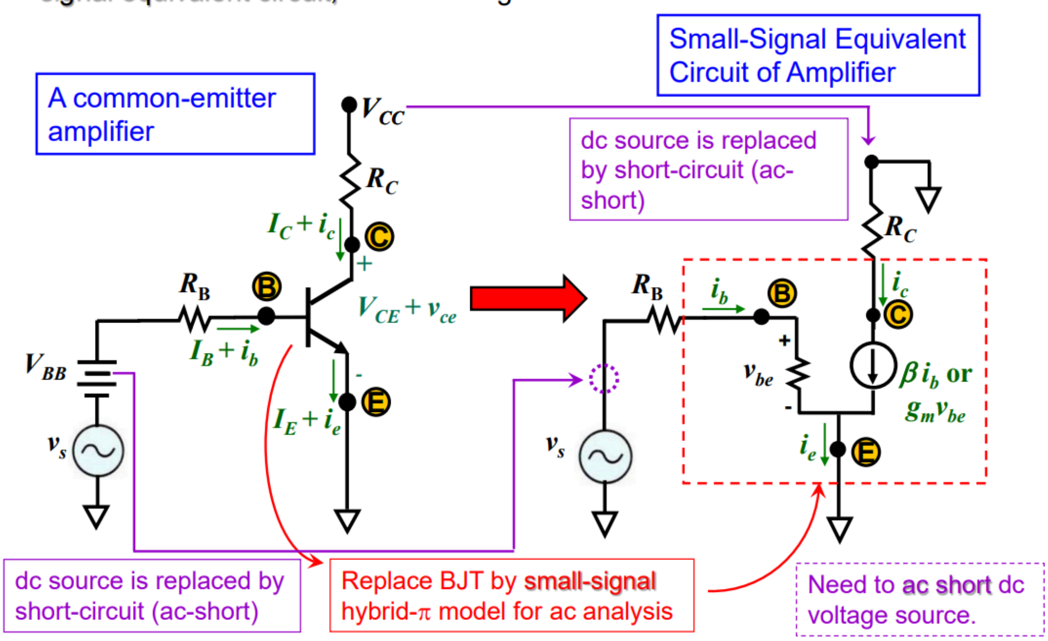
Full Analysis

<table>
<tr><td>❑ The dc bias point of the amplifier can be determined using the large-signal equivalent circuit, in the absence of the ac (small signal) source, v_s.

</td><td>Turn off AC source and perform DC analysis to obtain DC biassing point

I_C is used to find the small signal parameters</td></tr>
<tr><td>❑ The small-signal ac analysis of the amplifier is carried out using the small-signal equivalent circuit, and dc voltage sources need to be ac short.

</td><td>Turn off DC source, this includes the V_{CC}

Perform AC analysis</td></tr>
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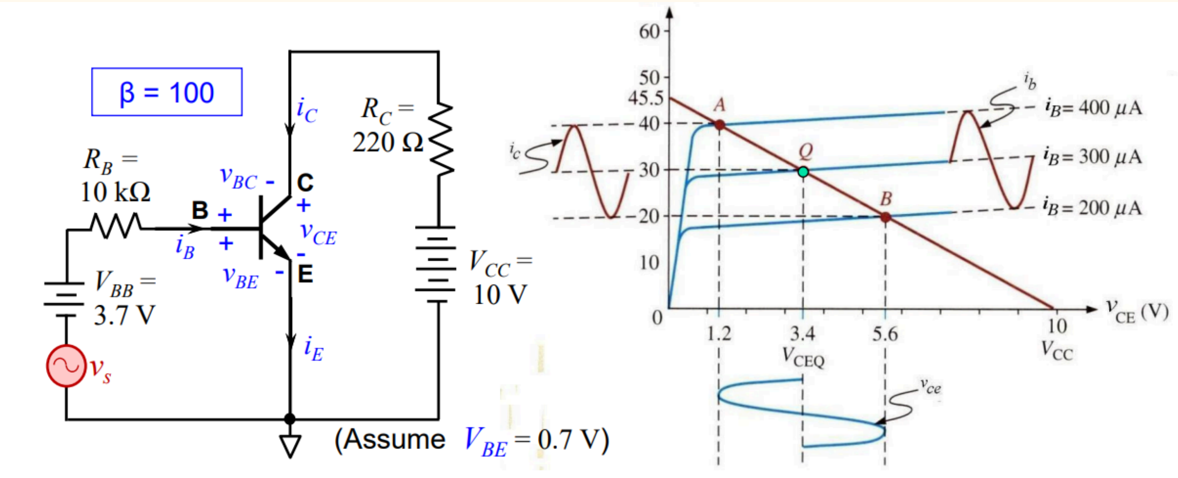
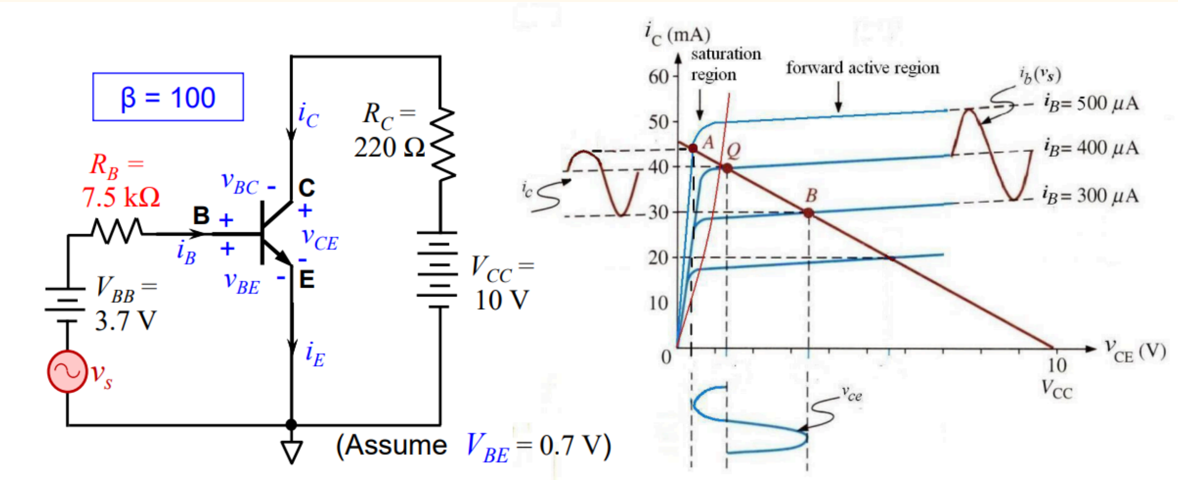
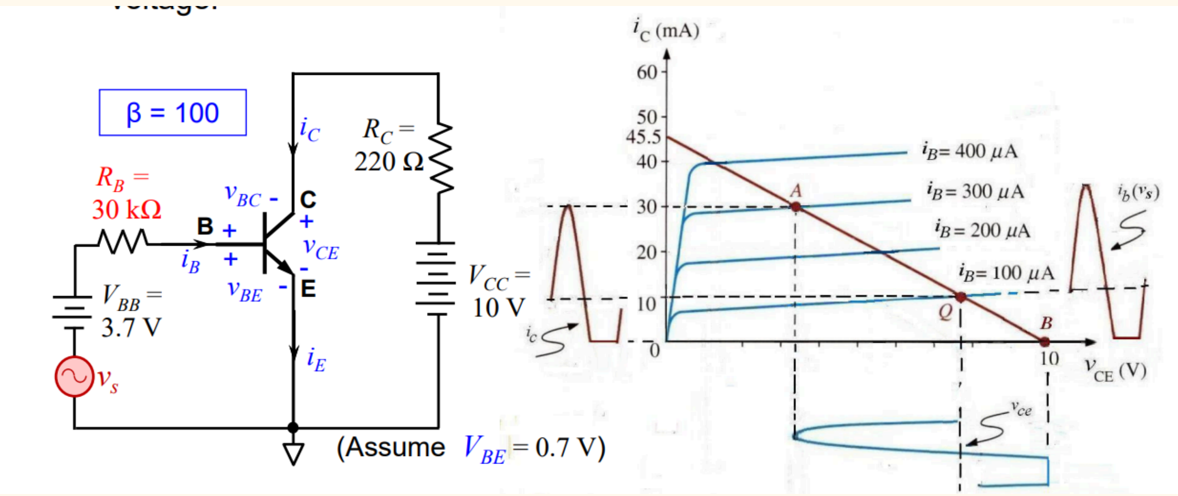
Art of Amplifier Biasing

<table>
<tr>
<td>
</td>
<td>Correct Biasing
No distortion from clipping or non linearity</td>
</tr>
<tr>
<td>
</td>
<td>Biasing Too High
Distortion from non linearity due to entering saturation region</td>
</tr>
<tr>
<td>
</td>
<td>Biasing Too Low
Distortion from clipping due to entering cut-off region</td>
</tr>
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